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Machine Learning-Based System for Cardiopulmonary Sound
Separation

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Abstract

Cardiopulmonary sound recordings often contain heart sounds, lung sounds, sensor contact noise, and environmental artefacts in the same audio signal. This overlap makes it difficult to process the signals accurately using simple filtering or traditional signal processing alone. This report presents a Final Year Project titled Machine Learning-Based System for Cardiopulmonary Sound Separation. The project aims to design and develop a software prototype that accepts mixed cardiopulmonary audio, applies suitable preprocessing, separates the signal into heart sound and lung sound outputs, and evaluates the separation quality using appropriate datasets and metrics.

The methodology follows an application-based prototype development approach. The literature review examines cardiopulmonary sound characteristics, traditional signal processing methods, machine learning and deep learning methods, available datasets, evaluation metrics, and prototype-related gaps. The proposed system uses public datasets, with HLS-CMDS identified as a relevant candidate because it provides mixed heart-lung recordings and corresponding source signals. The expected output is a reusable proof-of-concept software system that demonstrates heart-lung sound separation through audio preprocessing, model-based separation, output management, and planned evaluation. The project does not focus on disease diagnosis or clinical decision-making; its contribution is a structured software workflow for cardiopulmonary sound separation research and experimentation.

Table of Contents

Copyright	iii
Declaration	iv
Acknowledgements	v
Abstract	vi
Table of Contents.....	vii
List of Tables.....	x
List of Figures	xi
List of Abbreviations/Symbols.....	xii
List of Appendices	xiii
1. Chapter 1: Introduction	Error! Bookmark not defined.
1.1 Background.....	1
1.2 Problem Statement	1
1.3 Project Aim.....	2
1.4 Project Objectives	2
1.5 Project Scope.....	2
1.6 Project Significance.....	3
1.7 Expected Output	4
1.8 Report Organization	4
2. Chapter 2: Literature Review	6
2.1 Introduction	6
2.2 Literature Search and Screening Process.....	6
2.3 PRISMA Flow Diagram	7
2.4 Literature Review Matrix Summary	7
2.5 Cardiopulmonary Sound Characteristics.....	9
2.6 Heart and Lung Sound Separation Problem	9
2.7 Traditional Signal Processing Methods	10
2.8 Machine Learning and Deep Learning Methods	11
2.9 Datasets for Cardiopulmonary Sound Separation.....	12
2.10 Evaluation Metrics.....	13

2.11 Existing Systems and Prototype Relevance	14
2.12 Research Gaps	15
2.13 Summary	16
3. Chapter 3: Methodology	18
3.1 Introduction	18
3.2 Research / Development Approach.....	18
3.3 System Development Methodology.....	19
3.4 Dataset Selection and Preparation.....	20
3.5 Audio Preprocessing	21
3.6 Feature Extraction / Input Representation.....	22
3.7 Machine Learning Model Approach.....	22
3.8 System Architecture Overview	23
3.9 System Modules.....	23
3.10 Evaluation Methodology	24
3.11 Tools and Technologies	25
3.12 Chapter Summary	26
4. Chapter 4: Design and Implementation	27
4.1 Introduction	27
4.2 System Overview	27
4.3 System Architecture	27
4.4 Main System Modules.....	29
4.5 Database and Storage Design	30
4.6 Audio Upload and Preprocessing Flow	31
4.7 Machine Learning Model Integration	31
4.8 Heart and Lung Sound Output Generation.....	31
4.9 Result Management.....	32
4.10 User Interface Design	32
4.11 Security, Validation, and Error Handling	33
4.12 Implementation Status	33
4.13 Chapter Summary	34
5. Chapter 5: Testing and Evaluation	35

5.1 Introduction	35
5.2 Testing Strategy	35
5.3 Functional Testing	35
5.4 Dataset-Based Testing	36
5.5 Audio Separation Evaluation Metrics.....	37
5.6 Evaluation Procedure.....	38
5.7 Prototype / Usability Testing.....	38
5.8 Limitations of Testing.....	39
5.9 Summary.....	39
References	xiv
Appendix A: Full Literature Review Matrix	xviii
Appendix B: PRISMA Screening Summary	xxii
Appendix C: System Design Diagrams.....	xxiii
Appendix D: Planned Test Cases and Evaluation Metrics	xxiv

List of Tables

Table 2.1: Summary literature review matrix for key cardiopulmonary sound separation studies	8
No table of figures entries found.	30
Table 4.2: Planned database and storage records	30
Table 5.1: Planned functional test cases	35
Table 5.2: Planned audio separation evaluation metrics	37
Table 5.3: Full literature review matrix for selected Chapter 2 papers	xviii
Table 5.4: PRISMA screening summary for the literature review	xxii
Table 5.5: System design diagram summary	xxiii
Table 5.6: Appendix planned test cases and evaluation checks	xxiv

List of Figures

Figure 2.1: PRISMA flow diagram for the literature search and screening process. 7

Figure 4.1: Use case diagram for the cardiopulmonary sound separation system. 27

Figure 4.2: Component diagram for the cardiopulmonary sound separation system. 29

Figure 4.3: Audio processing workflow for the proposed system. 31

Figure 4.4: Sequence diagram for audio upload and cardiopulmonary sound separation. 32

List of Abbreviations/Symbols

Abbreviation	Description
FYP	Final Year Project
ML	Machine Learning
DL	Deep Learning
PCG	Phonocardiogram
HLS-CMDS	Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope
SNR	Signal-to-Noise Ratio
SDR	Signal-to-Distortion Ratio
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio
STFT	Short-Time Fourier Transform
NMF	Non-negative Matrix Factorization
VMD	Variational Mode Decomposition
CNN	Convolutional Neural Network
PESQ	Perceptual Evaluation of Speech Quality
STOI	Short-Time Objective Intelligibility
API	Application Programming Interface
UI	User Interface
UML	Unified Modeling Language
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
DOCX	Microsoft Word Open XML Document

List of Appendices

Appendix A: Full Literature Review Matrix	xviii
Appendix B: PRISMA Screening Summary	xxii
Appendix C: System Design Diagrams	xxiii
Appendix D: Planned Test Cases and Evaluation Metrics	xxiv

CHAPTER 1: INTRODUCTION

1.1 Background

Cardiopulmonary sounds are acoustic signals produced by heart activity and respiratory airflow. In practical auscultation, these sounds are commonly recorded from the chest using a stethoscope or digital stethoscope. Because the heart and lungs are close to each other anatomically and their acoustic components can overlap in a recording, a single chest sound signal may contain both heart sounds and lung sounds at the same time. This creates a signal separation problem: the system must estimate useful heart-related and lung-related components from a mixed cardiopulmonary recording.

Recent literature shows that cardiopulmonary sound processing is moving from purely traditional signal processing toward machine learning and deep learning approaches. Classical methods such as non-negative matrix factorization, blind filtering, and decomposition-based techniques remain useful because they are explainable and can work under limited-data conditions ([Grooby et al., 2023](#); [Fattahi et al., 2022](#)). More recent methods use deep learning architectures, phase-aware modeling, and hybrid approaches to improve single-channel chest sound separation ([Poh et al., 2024](#); [Wang et al., 2025](#)). These studies show that machine learning-based sound separation is a relevant research and software development direction.

This project is proposed as an application-based Software Engineering FYP. The focus is not to diagnose heart or lung disease. Instead, the focus is to design and develop a structured software prototype that can accept mixed cardiopulmonary audio, perform preprocessing, apply a machine learning-based separation method, and produce separated heart sound and lung sound outputs. Public datasets are important for this project because they allow the system to be tested and explained without relying on restricted clinical data. The HLS-CMDS dataset is especially relevant because it provides heart sounds, lung sounds, mixed recordings, and corresponding source signals recorded using a digital stethoscope and clinical manikin ([Torabi et al., 2025a](#)).

1.2 Problem Statement

The problem statements for this project are:

1. Noise and recording artefacts reduce the quality of cardiopulmonary sound recordings, making it difficult to process the signals accurately.
2. Heart and lung sounds overlap in time and frequency when recorded together, making it difficult to separate them using traditional signal processing methods.

3. Many existing studies focus on diagnosis or isolated algorithms, while fewer works provide a reusable end-to-end software prototype that performs sound separation and evaluates the output using suitable datasets and metrics.

1.3 Project Aim

The aim of this project is to design and develop a machine learning-based software prototype for cardiopulmonary sound separation that can process mixed chest sound recordings and produce separated heart sound and lung sound outputs.

1.4 Project Objectives

The objectives of this project are:

1. To study and apply suitable preprocessing techniques to reduce noise and improve the quality of cardiopulmonary sound recordings.
2. To design and implement a machine learning-based approach for separating mixed cardiopulmonary recordings into heart and lung sound outputs.
3. To develop a reusable software prototype and evaluate its separation performance using public datasets and suitable performance metrics.

The three objectives directly address the identified problems by focusing on signal quality improvement, heart-lung sound separation, and the development of an evaluated software prototype.

1.5 Project Scope

This project is limited to the development of a software prototype for cardiopulmonary sound separation. The system will focus on mixed audio recordings that contain heart and lung sound components and will produce two separated audio outputs: one representing the heart sound component and one representing the lung sound component.

The project scope includes:

1. Reviewing recent literature on cardiopulmonary sound separation, heart-lung sound separation, preprocessing, datasets, and evaluation metrics.
2. Using Python-based audio processing and machine learning tools to build the prototype.
3. Implementing preprocessing steps such as audio loading, normalization, segmentation or windowing where needed, and time-frequency representation where appropriate.

4. Implementing or adapting a machine learning-based separation approach suitable for the available dataset and project timeframe.
5. Using public datasets for training, testing, or evaluation, with attention to whether the dataset provides mixed and source audio signals.
6. Producing separated heart and lung sound files as the main system output.
7. Evaluating system performance using suitable metrics such as reconstruction quality, signal error, or source separation measures where reference signals are available.

The project scope excludes:

1. Medical diagnosis, disease detection, or clinical decision-making.
2. Hardware design or development of a physical digital stethoscope.
3. Claims that the system is clinically validated for patient use.
4. Use of private clinical datasets that cannot be accessed or cited appropriately.
5. Deployment as a production medical device.

The main constraints are the two-trimester FYP timeframe, dependence on public dataset availability, and the performance limitations of the selected machine learning method. Since dataset availability strongly affects evaluation, the project will avoid overclaiming generalizability beyond the data used for testing.

1.6 Project Significance

The significance of this project is mainly in software prototyping and biomedical audio processing. A working cardiopulmonary sound separation system can demonstrate how mixed chest sound recordings may be processed into separate heart and lung audio outputs. This can support future research workflows where separated audio is useful for listening, visualization, preprocessing experiments, or later model development.

From a Software Engineering perspective, the project is significant because it converts the separation problem into a structured application workflow. Instead of producing only an isolated algorithm script, the project aims to develop a reusable prototype with clear input handling, preprocessing, model inference, output storage, and evaluation. This is aligned with the proposal's focus on an application-based prototype/proof of concept.

From a research perspective, the project is significant because recent literature identifies challenges in dataset availability, method selection, evaluation, and system integration for cardiopulmonary sound separation ([Poh et al., 2024](#); [Wang et al., 2025](#); [Torabi et al., 2025a](#)). The project does not claim to solve these challenges completely. Its contribution is to build a feasible student-level system that demonstrates a complete separation workflow and provides a foundation for future improvement.

1.7 Expected Output

The expected output of this project is a functional software prototype for cardiopulmonary sound separation. The prototype is expected to:

1. Accept a mixed cardiopulmonary audio input.
2. Apply preprocessing steps required for the selected separation method.
3. Use a machine learning-based model or model-assisted pipeline to separate the input signal.
4. Produce two separated audio outputs: a heart sound output and a lung sound output.
5. Store or present the separated outputs so they can be reviewed by the user.
6. Report suitable evaluation results where reference signals or comparison data are available.

The final output is intended to be a proof of concept. It should demonstrate the feasibility of the proposed software workflow, but it should not be presented as a clinically approved diagnostic system.

1.8 Report Organization

This report is organized into five chapters.

Chapter 1 introduces the project background, problem statement, aim, objectives, scope, significance, expected output, and report organization.

Chapter 2 reviews recent literature related to cardiopulmonary sound characteristics, heart-lung sound separation, traditional signal processing methods, machine learning and deep learning methods, datasets, evaluation metrics, existing systems, and research gaps.

Chapter 3 presents the project methodology, including the development approach, data sources, preprocessing workflow, model selection, and evaluation plan.

Chapter 4 describes the system design and implementation direction, including the software architecture, major modules, data flow, storage design, model integration, and user interaction design for the prototype.

Chapter 5 presents the testing and evaluation plan, including functional testing, dataset-based testing, audio separation metrics, prototype usability testing, testing limitations, and work to be completed in the next project phase.

2. Chapter 2: Literature Review

2.1 Introduction

Cardiopulmonary auscultation produces acoustic recordings in which heart activity, respiratory airflow, sensor contact noise, and environmental interference may appear together in the same signal. In digital stethoscope recordings, these components are usually captured as a mixed audio waveform rather than as isolated heart and lung sources. This makes cardiopulmonary sound processing a source separation problem: the system must estimate heart-related and lung-related components from a combined recording before the separated outputs can be listened to, visualized, stored, or evaluated.

Recent work shows that the field is developing along two connected lines. The first line uses signal processing and decomposition methods such as non-negative matrix factorization (NMF), blind filtering, singular spectrum analysis, and variational mode decomposition (VMD). These approaches remain important because they are relatively explainable and can be useful when training data are limited ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Ullah and Zhang, 2024](#); [Han and Quan, 2025](#)). The second line uses learning-based methods, including U-Net-style models, attention networks, phase-aware transformers, and reprogrammed speech models ([Poh et al., 2024](#); [Zhang et al., 2025](#); [Wang et al., 2025](#); [Poh et al., 2026](#)). The contrast between these lines is central to this FYP: classical methods offer interpretability and simpler implementation, while deep learning methods offer stronger modeling capacity but usually require more data, computation, and careful evaluation.

This chapter reviews the literature thematically. It first discusses cardiopulmonary sound characteristics and the heart-lung separation problem. It then compares traditional signal processing methods, machine learning and deep learning methods, datasets, evaluation metrics, and existing system or prototype evidence. Related heart sound and lung sound papers are included only where they support separation, preprocessing, dataset selection, or evaluation. Disease detection and classification papers are treated as background evidence, not as the objective of this FYP ([Roy et al., 2024](#); [Zhao et al., 2024](#)).

2.2 Literature Search and Screening Process

The literature review process followed a structured screening approach to ensure that the selected studies were relevant to cardiopulmonary sound separation, audio preprocessing, public datasets, machine learning methods, evaluation metrics, and software prototype development. A total of 96 records were considered. After screening and access checking, 47 records were excluded because they were outside the project scope, duplicated earlier

records, weakly related to sound separation, outside the target review period, or not sufficiently accessible for use as supporting evidence.

After screening, 49 records remained eligible for the review pool. From the eligible records, 35 papers were selected and used in the Chapter 2 synthesis. These consisted of 18 core studies directly related to cardiopulmonary, heart-lung, or chest sound separation, and 17 background/supplementary studies related to datasets, preprocessing, machine learning methods, evaluation metrics, or supporting context. The remaining 14 eligible records were retained as backup evidence but were not cited in the current report. Excluded records were not used as supporting evidence in the literature synthesis.

The PRISMA flow diagram in [Figure 2.1](#) summarizes the search and screening process in a reader-friendly form.

2.3 PRISMA Flow Diagram

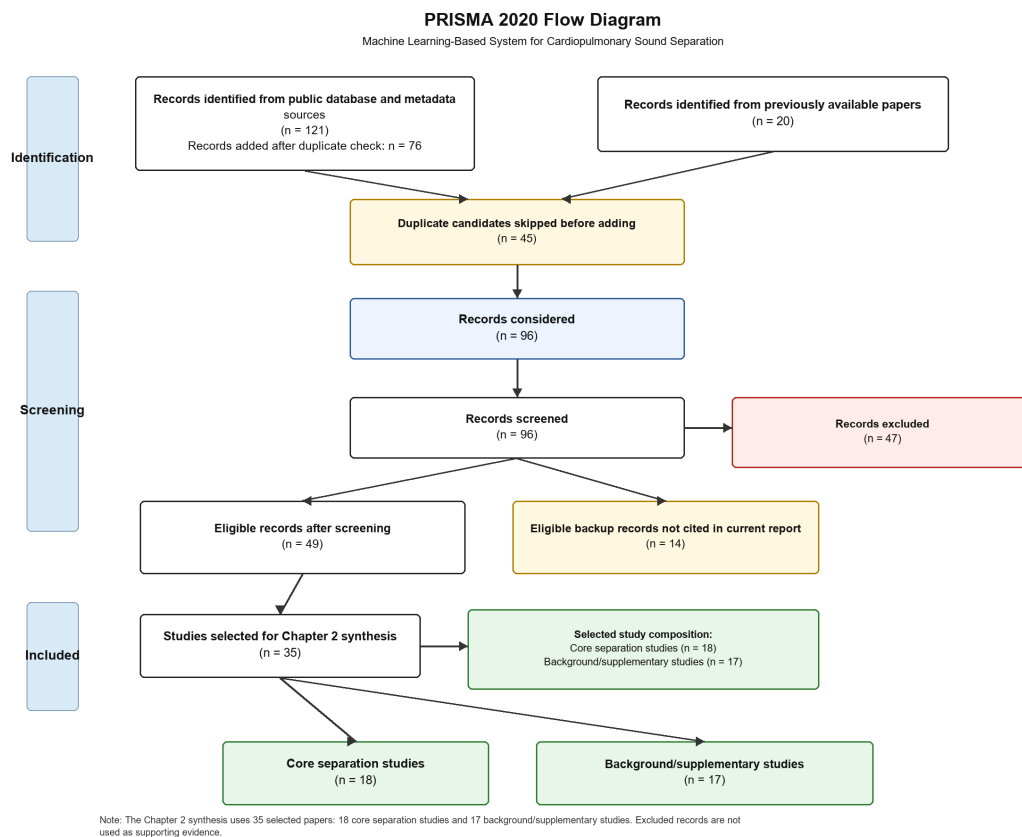


Figure 2.1: PRISMA flow diagram for the literature search and screening process.

2.4 Literature Review Matrix Summary

The literature review matrix summarizes the main studies reviewed in this chapter. A concise version is presented below, while the complete matrix is provided in Appendix A.

Table 2.1: Summary literature review matrix for key cardiopulmonary sound separation studies

No.	Author(s) & Year	Focus	Method / Model	Dataset	Key Finding	Relevance to This Project
1	Grooby et al. (2023) (Grooby et al., 2023)	Neonatal chest sound separation	NMF and NMCF	Neonatal mixtures and noisy chest sounds	Classical decomposition can improve heart and lung sound quality.	Provides a baseline for separating mixed cardiopulmonary recordings.
2	Fattahi et al. (2022) (Fattahi et al., 2022)	Noisy chest sound filtering	Wavelet transform, PCA, periodic component analysis, SOBI	Neonatal chest sound recordings	Blind filtering improves signal quality under noisy conditions.	Supports preprocessing and traditional baseline design.
3	Poh et al. (2024) (Poh et al., 2024)	Real-time neonatal separation	1D encoder-decoder with transformer mask generation	Neonatal chest recordings	Deep learning can support real-time heart-lung separation.	Main model reference for a machine learning-based prototype.
4	Sun et al. (2024) (Sun et al., 2024)	Hybrid heart-lung separation	DAE, NMF, and VMD	Heart-lung mixtures	Hybrid decomposition and learning methods improve separation quality.	Shows how preprocessing and ML can be combined.
5	Ullah and Zhang (2024) (Ullah and Zhang, 2024)	Data-constrained single-channel separation	MCRA, OMLSA, and NMF-HMM	Heart and lung sound mixtures	Unsupervised methods can be useful when labelled data are limited.	Supports feasible low-data separation choices.
6	Wang et al. (2023) (Wang et al., 2023a)	Robust heart-lung separation	SNMF-DCNN-MCUNet	Heart-lung mixtures with different lung sound types	U-Net-style models can improve standard separation metrics.	Guides model architecture and evaluation planning.
7	Zhang et al. (2025) (Zhang et al., 2025)	Attention-based separation	Joint attention and semi-supervised network	Simulated and clinical cardiopulmonary recordings	Attention and semi-supervised learning improve recent separation models.	Defines a recent advanced method for comparison.
8	Wang et al. (2025) (Wang et al., 2025)	Phase-aware chest sound separation	Dual-path transformer with phase enhancement	Constructed chest sound mixtures	Phase information can affect reconstructed sound quality.	Supports time-frequency and reconstruction design choices.
9	Torabi et al. (2025) (Torabi et al., 2025a)	Public separation dataset	HLS-CMDS dataset descriptor	Manikin heart, lung, mixed, and source recordings	Provides paired mixed and source recordings for evaluation.	Main public dataset candidate for the prototype.
10	Tsai et al. (2023) (Tsai et al., 2023)	Feasible deep separation pipeline	Deep network-based model	Public heart/lung sounds used to generate mixtures	A simpler neural model can separate synthetic heart-lung mixtures.	Useful for realistic student-level implementation planning.

2.5 Cardiopulmonary Sound Characteristics

Heart and lung sounds differ in their physiological origin and acoustic pattern, but they are not cleanly separated in practical chest recordings. Heart sounds are commonly described using phonocardiogram (PCG) concepts such as S1, S2, cycles, envelopes, onset/offset timing, and time-frequency representations. Lung sounds are linked to respiratory airflow and breathing phase, and may contain continuous or discontinuous events depending on the recording context. Reviews of PCG signal analysis emphasize that preprocessing, segmentation, feature extraction, and noise handling are necessary before reliable higher-level analysis can be attempted ([Roy et al., 2024](#); [Zhao et al., 2024](#)).

For this FYP, the most important characteristic is overlap. Heart components can interfere with lung sound analysis, and lung or breathing components can interfere with heart sound analysis. Denoising studies make this point indirectly: heart-sound denoising work treats breathing, lung, and hospital noise as contaminants that reduce PCG quality, while recent lung-sound denoising studies show that respiratory recordings also require robust noise suppression and representation before downstream processing ([Ali et al., 2023](#); [Jakubec et al., 2025](#); [Shuvo et al., 2026](#); [Behera et al., 2026](#); [Pouyani et al., 2022](#)). These papers are not all direct heart-lung separation papers, but they support the need for preprocessing in any practical cardiopulmonary audio pipeline.

Temporal structure is also important. Heart sounds have a stronger quasi-periodic structure than many respiratory components, while lung sounds follow respiratory phase and airflow variation. This difference motivates approaches that use periodicity, phase, or segmentation information to support separation or evaluation ([Torabi et al., 2023](#); [Somarathne et al., 2026](#); [Pal and Barney, 2024](#)). Input quality is another practical factor. Neonatal and pediatric auscultation studies show that recording quality can vary significantly, which makes signal quality assessment relevant before or alongside separation ([Grooby et al., 2022](#); [Arjouni et al., 2023](#)). Therefore, a cardiopulmonary sound separation system should be designed as an audio-processing workflow with input checking, preprocessing, separation, and output inspection, rather than as a direct clinical decision system.

2.6 Heart and Lung Sound Separation Problem

The core technical problem is to estimate heart and lung components from a mixed cardiopulmonary recording. This is challenging because the mixture is often single-channel, the sources overlap in time and frequency, and clean reference sources are rarely available in real clinical recordings. Neonatal chest sound separation studies illustrate this difficulty clearly: recordings may include heart activity, lung sounds, respiratory support noise, movement, and sensor

noise, so separation methods must work under noisy and underdetermined conditions ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Poh et al., 2024](#)).

Early and traditional separation work often treats the mixture as a signal decomposition problem. NMF and NMCF methods model spectrogram structure through basis patterns and activations, while blind filtering methods attempt to suppress unwanted components without requiring a large supervised dataset ([Grooby et al., 2023](#); [Fattahi et al., 2022](#)). These approaches are useful baselines because they make the separation assumptions visible. However, they can be sensitive to parameter choices and mixture assumptions.

Recent papers move the separation problem toward data-driven and hybrid models. NeoSSNet is important because it frames neonatal chest sound separation as a real-time deep learning task, which links algorithm design to deployable processing ([Poh et al., 2024](#)). Phase-enhanced transformer work further argues that single-channel chest sound separation should consider phase reconstruction, not only magnitude reconstruction ([Wang et al., 2025](#)). JASSNet and reprogrammed automatic speech recognition models represent more advanced learning-based directions, using attention, semi-supervision, or transfer from pretrained speech models ([Zhang et al., 2025](#); [Poh et al., 2026](#)). These studies show the direction of current research, but their complexity also creates a feasibility issue for an FYP1 prototype.

Some studies use cardiopulmonary separation mainly as a preprocessing stage before respiratory sound identification or classification ([Zheng and Sanei, 2024](#); [Zheng et al., 2025](#)). These papers are relevant because they show that separation can improve or support later audio analysis, but they should not shift the project aim toward diagnosis. This FYP remains focused on separating and presenting heart and lung sound components.

2.7 Traditional Signal Processing Methods

Traditional signal processing methods provide the foundation for heart-lung sound separation because they exploit structure in the signal itself. NMF-based methods are among the clearest examples. In neonatal chest sound separation, NMF and NMCF provide interpretable component estimates for heart and lung extraction ([Grooby et al., 2023](#)). Periodicity-guided NMF extends this idea by using differences between cardiovascular and respiratory periodicity to guide blind source separation ([Torabi et al., 2023](#)). LingoNMF later adds language-model feedback for NMF parameter tuning, which is conceptually useful but introduces an additional system layer that may be unnecessary for a first FYP prototype ([Torabi et al., 2025b](#)).

Other recent work combines NMF with learning-based components. Heart-lung sound separation by NMF and deep learning uses decomposition ideas together with neural modeling, while SNMF-DCNN-MCUNet combines sparse

NMF-derived information with a U-Net-style architecture ([Wang et al., 2023b](#); [Wang et al., 2023a](#)). These papers are useful for comparison because they occupy the middle ground between transparent decomposition and fully learned separation. For this FYP, such hybrid methods are attractive because they suggest a practical design pattern: use preprocessing or decomposition to simplify the input before applying a learned model.

VMD and related decomposition methods form another important method family. DAE-NMF-VMD combines a deep autoencoder, NMF, and VMD for heart and lung sound separation, making it a key hybrid reference ([Sun et al., 2024](#)). GIHO-VMD applies optimized VMD to lung sound separation and provides a recent decomposition-based comparator ([Zhang et al., 2026](#)). Compared with NMF, VMD-style approaches focus more directly on decomposing a waveform into modes, but they may still depend on parameter selection, signal assumptions, and the way modes are selected or recombined.

Blind filtering, singular spectrum analysis, and respiratory preprocessing methods further broaden the traditional-method landscape. Blind filtering for noisy neonatal chest sounds is relevant because it addresses realistic noise before later deep learning work ([Fattahi et al., 2022](#)). Singular spectrum analysis offers a recent direct cardiorespiratory separation candidate, reinforcing that non-deep-learning methods remain active in recent literature ([Han and Quan, 2025](#)). Harmonic-percussive source separation and empirical mode decomposition have also been used for respiratory sound segmentation, showing how source-separation ideas can support lung-sound preprocessing even when the task is not paired heart-lung separation ([Rocha et al., 2023](#)).

Overall, traditional methods are valuable for FYP1 because they are easier to explain, can act as baselines, and clarify the assumptions behind separation. Their limitations are also clear: they can be sensitive to noise, parameter tuning, and dataset conditions. Therefore, they are best treated as baselines or components in a broader pipeline rather than as the only solution direction.

2.8 Machine Learning and Deep Learning Methods

Machine learning and deep learning methods are increasingly used because they can learn nonlinear mappings from mixed or noisy audio to cleaner outputs. U-Net-style encoder-decoder models are especially relevant because they have been used for denoising and source separation tasks. NeoSSNet is the strongest direct anchor for this FYP because it targets real-time neonatal chest sound separation and connects model design with a practical processing goal ([Poh et al., 2024](#)). A simpler deep network-based heart and lung sound separation study is also useful because it shows a more feasible supervised learning pipeline, although its synthetic-mixture setting limits how strongly it can support real-world claims ([Tsai et al., 2023](#)).

Hybrid neural methods show how classical signal-processing assumptions can be combined with learned representations. DAE-NMF-VMD integrates autoencoding with NMF and VMD, while SNMF-DCNN-MCUNet uses sparse NMF-derived information within a U-Net framework ([Sun et al., 2024](#); [Wang et al., 2023a](#)). Compared with purely classical methods, these approaches can model more complex mixture patterns. Compared with fully end-to-end deep models, they retain a clearer link to signal decomposition. This makes hybrid methods especially relevant for a student project that must be explainable as well as functional.

Attention, semi-supervised learning, phase-aware modeling, and pretrained-model adaptation represent the more advanced end of the current literature. JASSNet uses joint attention and semi-supervised learning for heart and lung sound separation ([Zhang et al., 2025](#)). A phase-enhanced dual-path transformer emphasizes that phase information can affect the quality of single-channel chest sound reconstruction ([Wang et al., 2025](#)). Reprogramming automatic speech recognition models for neonatal chest sound separation explores whether pretrained speech models can be adapted to biomedical audio ([Poh et al., 2026](#)). These works are valuable for identifying the state of the field, but they also define a boundary for this FYP: reproducing complex transformer or foundation-model systems may be beyond a realistic FYP1 implementation unless the scope is narrowed carefully.

Related denoising and preprocessing models support the separation workflow. LU-Net, robust PCG denoising, self-attention lung denoising, DWT-UNet respiratory denoising, and DWT-ANN lung denoising show that deep learning is frequently used to improve signal quality before further analysis ([Ali et al., 2023](#); [Jakubec et al., 2025](#); [Shuvo et al., 2026](#); [Behera et al., 2026](#); [Pouyani et al., 2022](#)). These studies should not be described as direct cardiopulmonary separation solutions unless they actually separate heart and lung sources. Their role in this chapter is to justify preprocessing, robustness, and noise-handling decisions in the proposed system.

The comparison across methods suggests a practical design choice. A feasible FYP implementation should not simply choose the most complex recent model. It should select a method that can be implemented, explained, tested on available data, and integrated into a working software pipeline. In this sense, direct separation models such as NeoSSNet, hybrid methods such as DAE-NMF-VMD or NMF-plus-deep-learning models, and simpler supervised separation networks provide more practical guidance than diagnosis-only or highly specialized architectures.

2.9 Datasets for Cardiopulmonary Sound Separation

Dataset availability is one of the main constraints in cardiopulmonary sound separation. Many available heart sound or respiratory sound datasets are designed for classification, segmentation, or benchmarking rather than source separation. Reviews of PCG and heart-sound deep learning literature are useful for understanding this dataset landscape, but they also show that dataset purpose matters: a dataset suitable for murmur detection or cardiac classification is not automatically suitable for supervised heart-lung separation ([Roy et al., 2024](#); [Zhao et al., 2024](#)).

The most relevant dataset for this FYP is HLS-CMDS. The HLS-CMDS descriptor paper reports a heart and lung sounds dataset recorded from a clinical manikin using a digital stethoscope, with individual heart sounds, individual lung sounds, mixed recordings, and corresponding source signals for mixed recordings ([Torabi et al., 2025a](#)). This makes it more directly useful for training, testing, or evaluating source separation than datasets containing only labels or disease categories. Its limitation is also important: because it is based on a clinical manikin, it should not be overclaimed as fully representative of real patient variability.

Other dataset papers support the surrounding context but should be positioned carefully. The BUET multi-disease heart sound dataset provides a public auscultation resource, but it is primarily a heart-sound diagnosis dataset rather than a paired heart-lung separation dataset ([Ali et al., 2026](#)). The George B. Moody PhysioNet Challenge 2022 paper documents an important PCG benchmark, but its task is murmur detection, not source separation ([Reyna et al., 2023](#)). Sound-Dr provides respiratory audio dataset and baseline AI context, but it also does not provide paired heart and lung source targets for separation ([Hoang et al., 2023](#)).

This distinction leads to a clear dataset strategy for the FYP. HLS-CMDS should be treated as the primary separation-relevant dataset candidate because it includes mixed and source signals. PCG and respiratory datasets should be used only as background, preprocessing, or auxiliary evaluation context unless they can be adapted without changing the project objective. The chapter should keep this distinction explicit so the report does not blur source separation with disease classification.

2.10 Evaluation Metrics

Evaluation should match the objective of the system. Since this FYP is about separating cardiopulmonary sound components, the main evaluation should focus on signal reconstruction, source quality, and processing behavior rather than diagnostic accuracy. Separation papers commonly report objective signal-level or audio-level comparisons, depending on whether reference source signals are available ([Grooby et al., 2023](#); [Sun et al., 2024](#); [Wang et al., 2023a](#);

[Wang et al., 2023b](#)). When clean source references are available, evaluation can be more direct. When only mixed or clinical recordings are available, evaluation becomes less certain and may require indirect quality measures.

Recent work also shows that evaluation is not only about amplitude or magnitude. Phase-enhanced single-channel chest sound separation emphasizes that phase reconstruction can influence output quality, especially when spectrogram-based models are converted back into waveforms ([Wang et al., 2025](#)). VMD-based lung sound separation and other decomposition methods further show that method comparison depends on the choice of reconstruction and error measures ([Zhang et al., 2026](#)). For this FYP, the evaluation plan should therefore avoid relying on a single number if multiple output qualities matter.

Segmentation and quality assessment provide secondary evaluation support. Heart sound onset/offset detection can help assess timing and segmentation quality, while breathing phase identification can support respiratory signal interpretation ([Somarathne et al., 2026](#); [Pal and Barney, 2024](#)). Real-time neonatal heart and lung sound quality assessment also suggests that input quality should be considered before judging the separation result ([Grooby et al., 2022](#)). These papers support practical system checks, but they should be separated from source-separation metrics in the report.

Classification metrics require special caution. Some studies evaluate separation through downstream respiratory classification or cardiopulmonary analysis ([Zheng and Sanei, 2024](#); [Zheng et al., 2025](#)). The PhysioNet murmur challenge is also important for benchmark context, but it is not a separation benchmark ([Reyna et al., 2023](#)). For this project, classification accuracy can be mentioned as background only. It should not be used as the main claim of success unless the methodology explicitly includes a justified downstream task.

2.11 Existing Systems and Prototype Relevance

The FYP is not only a model-selection exercise; it is a software prototype. Therefore, system-oriented literature is useful even when it does not directly solve heart-lung separation. StethAid shows how digital auscultation can be organized as a pediatric software platform, while TinyML-based digital stethoscope work shows how organ-sound processing can be linked to embedded or constrained deployment contexts ([Arjoun et al., 2023](#); [Mutlu et al., 2024](#)). These papers support the need for a usable workflow, but this FYP should remain a software-based separation system rather than a hardware or embedded-device project.

Real-time and robustness requirements also shape the prototype. NeoSSNet provides a direct example of real-time neonatal chest sound separation ([Poh et al., 2024](#)). Heart sound and lung sound denoising studies show that noisy inputs

and real-time processing concerns appear throughout biomedical audio work ([Ali et al., 2023](#); [Shuvo et al., 2026](#)). Compressed sensing and CNN-based reconstruction of acoustic cardiopulmonary signals adds another implementation perspective: recording and reconstruction quality can influence the usefulness of downstream processing ([Baeyens et al., 2025](#)).

For this FYP, the literature supports a modular system design. A realistic prototype should allow a user to upload or select an audio file, preprocess the waveform, apply a separation method, save separated heart and lung outputs, compute available signal-quality or separation metrics, and present the results through an interface. The prototype does not need to reproduce a full clinical auscultation platform. Its contribution should be a clear, reproducible, and explainable pipeline for cardiopulmonary sound separation.

2.12 Research Gaps

The reviewed literature identifies four gaps that justify this FYP.

First, there is a dataset gap. Many strong separation studies use private, neonatal, clinical, simulated, or otherwise difficult-to-reproduce datasets ([Poh et al., 2024](#); [Poh et al., 2026](#); [Wang et al., 2025](#)). HLS-CMDS helps address this gap because it provides mixed recordings and corresponding source signals, but it is still a manikin dataset and should not be treated as a complete substitute for real patient data ([Torabi et al., 2025a](#)). Meanwhile, several useful public datasets are classification-oriented and do not provide paired source signals ([Ali et al., 2026](#); [Hoang et al., 2023](#); [Reyna et al., 2023](#)).

Second, there is a method-selection gap for student-level implementation. Traditional methods such as NMF, VMD, blind filtering, and singular spectrum analysis are explainable and feasible, but they may be sensitive to parameter settings and mixture assumptions ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Torabi et al., 2023](#); [Zhang et al., 2026](#); [Han and Quan, 2025](#)). Advanced deep methods can model more complex patterns, but attention networks, transformers, and pretrained-model reprogramming may require more data, computation, and implementation effort than is realistic for FYP1 ([Zhang et al., 2025](#); [Wang et al., 2025](#); [Poh et al., 2026](#)). This creates a need for a carefully scoped ML-based method that balances feasibility, explainability, and separation quality.

Third, there is an evaluation gap. Separation papers use different datasets, metrics, and validation assumptions, while some related works judge separation mainly through downstream classification performance ([Zheng and Sanei, 2024](#); [Zheng et al., 2025](#)). For a sound separation FYP, this is not sufficient. The system should report separation-relevant metrics when reference signals are available and use preprocessing or quality-assessment evidence only as

supporting information ([Grooby et al., 2022](#); [Somarathne et al., 2026](#); [Pal and Barney, 2024](#)).

Fourth, there is a system-integration gap. Much of the literature focuses on algorithms, datasets, or classification pipelines, while a student FYP also needs a working software flow that handles audio input, preprocessing, separation, output storage, and user-facing review. Denoising, phase reconstruction, segmentation, and acquisition quality all affect the usefulness of such a system ([Jakubec et al., 2025](#); [Wang et al., 2025](#); [Baeyens et al., 2025](#)). The proposed project can address this gap by building a practical ML-based cardiopulmonary sound separation system and evaluating it as an audio-processing prototype, not as a diagnostic tool.

Therefore, the research gap for this FYP can be stated as follows: recent literature provides promising methods and datasets for heart-lung or cardiopulmonary sound separation, but there remains a need for a clear, reproducible, student-feasible software system that applies an appropriate ML-based separation workflow, uses separation-relevant data where available, reports suitable signal-level evaluation, and avoids unsupported diagnostic claims.

2.13 Summary

This chapter reviewed recent 2022-2026 literature relevant to machine learning-based cardiopulmonary sound separation. Traditional methods such as NMF, VMD, blind filtering, and singular spectrum analysis provide explainable foundations and useful baselines ([Grooby et al., 2023](#); [Ullah and Zhang, 2024](#); [Fattahi et al., 2022](#); [Han and Quan, 2025](#)). Hybrid and deep learning methods, including U-Net-based models, attention networks, phase-aware transformers, and reprogrammed speech models, represent the main direction of recent separation research ([Poh et al., 2024](#); [Sun et al., 2024](#); [Wang et al., 2023a](#); [Zhang et al., 2025](#); [Wang et al., 2025](#); [Poh et al., 2026](#)).

The literature also shows that dataset selection and evaluation are major challenges. HLS-CMDS is the most relevant dataset candidate for this FYP because it provides mixed and source heart/lung recordings, while other heart sound and respiratory datasets are useful mainly for background or auxiliary context ([Torabi et al., 2025a](#); [Ali et al., 2026](#); [Hoang et al., 2023](#); [Reyna et al., 2023](#)). Evaluation should emphasize separation quality, reconstruction, signal quality, and practical processing behavior rather than disease classification.

Overall, the reviewed evidence supports a practical FYP system that preprocesses cardiopulmonary audio, separates heart and lung components using a feasible ML-based method, stores the separated outputs, and reports suitable evaluation information. The next chapter should translate this review into a methodology by selecting the data source, defining the processing

pipeline, choosing the separation model, and specifying the evaluation procedure for the prototype.

3. Chapter 3: Methodology

3.1 Introduction

This chapter describes the methodology proposed for developing the machine learning-based cardiopulmonary sound separation system. The project is an application-based Software Engineering project with a prototype/proof-of-concept contribution. Its purpose is to process mixed cardiopulmonary audio and produce two separated audio outputs: a heart sound output and a lung sound output.

The methodology is designed around the project objectives stated in Chapter 1. These objectives include studying existing techniques, designing a software system architecture, implementing preprocessing and machine learning-based separation modules, testing the system using suitable public datasets, and evaluating the separated outputs using appropriate metrics. The methodology does not include disease diagnosis, clinical decision-making, or hardware development.

The literature review in Chapter 2 shows that cardiopulmonary sound separation can be approached using traditional signal processing, hybrid signal-processing and machine-learning methods, and deep learning models. It also shows that dataset selection and evaluation are major issues because not all heart sound or lung sound datasets provide paired mixed and source signals. Therefore, this methodology emphasizes a reproducible software workflow, careful dataset selection, preprocessing, model-based separation, and signal-level evaluation.

3.2 Research / Development Approach

This project follows an applied research and prototype development approach. The research component is used to study cardiopulmonary sound separation methods, identify suitable datasets, and select appropriate evaluation metrics. The development component is used to design and implement a software prototype that applies these methods to mixed cardiopulmonary audio.

The approach is literature-informed rather than diagnosis-driven. Existing work shows that heart and lung sounds can overlap in chest recordings and that separation methods may use NMF, blind filtering, hybrid decomposition, U-Net-style networks, attention models, or phase-aware deep learning ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Poh et al., 2024](#); [Sun et al., 2024](#); [Wang et al., 2025](#)). However, this project will select a method based on feasibility, dataset availability, explainability, and suitability for integration into a student-level software system.

The expected development workflow is:

1. Review relevant literature and datasets.
2. Define system requirements and the separation workflow.
3. Prepare public audio datasets.
4. Implement preprocessing and input representation steps.
5. Implement or adapt a machine learning-based separation method.
6. Integrate the method into a software prototype.
7. Generate separated heart and lung sound outputs.
8. Evaluate the outputs using suitable separation or signal-quality metrics.
9. Document the results, limitations, and future improvements.

3.3 System Development Methodology

The project will use an iterative prototyping methodology. This methodology is suitable because the final system depends on dataset availability, preprocessing quality, model feasibility, and evaluation results. Iterative prototyping allows the project to begin with a simple working pipeline and then improve the preprocessing, model, and interface in stages.

The proposed development phases are:

Phase	Description	Expected output
Requirements analysis	Identify functional and non-functional requirements based on the proposal, literature review, and FYP scope.	Initial requirement list and system boundary.
Dataset preparation	Select and organize public cardiopulmonary audio datasets.	Prepared dataset folder structure and metadata notes.
Preprocessing prototype	Implement audio loading, normalization, segmentation/windowing, and time-frequency conversion where needed.	Reusable preprocessing module.
Model prototype	Implement or adapt a feasible machine learning-based separation model or model-assisted pipeline.	Initial separation module.
System integration	Connect input handling, preprocessing, separation, output saving, and evaluation steps.	Functional software prototype.
Testing and evaluation	Test the system using selected audio samples and evaluate separated outputs.	Evaluation results and limitations.
Documentation	Document design, methodology, testing, and future work for the FYP report.	Updated report chapters and final submission materials.

This approach supports the FYP1 requirement to demonstrate a clear development plan while leaving space for implementation refinement in the next phase.

3.4 Dataset Selection and Preparation

Public datasets will be used for training, testing, or evaluation. Dataset selection is important because a dataset designed for disease classification is not necessarily suitable for heart-lung source separation. For this project, the preferred dataset should contain mixed cardiopulmonary recordings and corresponding heart/lung source signals, because source signals allow direct evaluation of the separated outputs.

The dataset selection criteria are:

1. The dataset should contain audio recordings relevant to heart sounds, lung sounds, or mixed cardiopulmonary sounds.
2. The dataset should be publicly available or accessible without restricted clinical access.
3. The dataset should provide clear metadata, citation information, and usage conditions.
4. The dataset should preferably include mixed recordings and corresponding source signals.
5. The dataset format should be suitable for Python-based audio processing, preferably .wav.
6. The dataset should support training, testing, or evaluation within the FYP timeframe.

Based on the current dataset inventory, HLS-CMDS is the main dataset candidate for this project. The HLS-CMDS descriptor paper reports a dataset of heart and lung sounds recorded from a clinical manikin using a digital stethoscope, including heart sounds, lung sounds, mixed sounds, and source signals for mixed recordings ([Torabi et al., 2025a](#)). This makes it suitable as a possible training, testing, or evaluation source for cardiopulmonary sound separation. The dataset inventory records the related paper DOI as 10.1109/IEEEDATA.2025.3566012 and the dataset DOI/PID as 10.17632/8972jxbmp.

Other public datasets may be used only as supporting background or auxiliary testing if they are relevant and accessible. However, datasets that contain only heart sounds, only lung sounds, or classification labels will not be treated as direct paired source-separation datasets unless they can be adapted in a

justified way. This distinction is important because the project objective is separation, not disease classification.

The preparation process will include:

1. Recording the dataset source, citation, license or access status, and folder structure.
2. Inspecting audio file format, sampling rate, duration, and channel count.
3. Separating mixed recordings from source heart and lung recordings where available.
4. Creating a consistent naming and metadata structure for input mixtures and target outputs.
5. Splitting data into training, validation, and testing subsets if the selected model requires training.
6. Keeping notes on any excluded files, missing files, or unusable samples.

3.5 Audio Preprocessing

Audio preprocessing is required before model inference or training because raw recordings may differ in sampling rate, duration, amplitude, noise level, and channel format. The preprocessing stage will convert audio data into a consistent format suitable for the selected model.

The proposed preprocessing steps are:

1. Load audio files using a Python audio library.
2. Convert audio to mono if required by the selected model.
3. Resample audio to a consistent sampling rate.
4. Normalize amplitude to reduce scale differences between files.
5. Trim silence or pad short files where needed.
6. Segment long recordings into fixed-length windows if the model requires fixed input length.
7. Apply optional noise reduction or filtering only when justified by the selected method.
8. Save preprocessing metadata so that outputs can be traced back to the original inputs.

The preprocessing design will remain conservative. It should improve consistency without changing the purpose of the task or introducing unsupported assumptions. Since Chapter 2 shows that preprocessing and

signal quality affect separation outcomes, this stage will be treated as part of the system pipeline rather than as a minor helper step ([Roy et al., 2024](#); [Zhao et al., 2024](#)).

3.6 Feature Extraction / Input Representation

The input representation will depend on the final separation model. Two main options are considered.

The first option is waveform-based processing, where the model receives audio segments directly or through a learned representation. This can preserve time-domain information but may require more careful model design and training.

The second option is time-frequency processing, where the audio is converted into a spectrogram or related representation. This is common in audio separation because the model can learn patterns in frequency and time. A spectrogram-based system may estimate masks or separated representations for heart and lung components before reconstructing waveform outputs. If a time-frequency method is used, phase handling must be considered because phase reconstruction can affect waveform quality in single-channel separation ([Wang et al., 2025](#)).

For the first prototype, the input representation should be selected based on feasibility and compatibility with the chosen model. A practical approach is to begin with short audio windows and a time-frequency representation, then reconstruct separated heart and lung waveform outputs after model inference. The final choice will be documented after implementation testing.

3.7 Machine Learning Model Approach

The machine learning model will be selected to balance relevance, feasibility, explainability, and implementation effort. Chapter 2 identifies several possible method families, including NMF-based methods, hybrid decomposition and neural methods, U-Net-style deep learning models, attention-based methods, and phase-aware transformer models ([Grooby et al., 2023](#); [Wang et al., 2023a](#); [Zhang et al., 2025](#); [Wang et al., 2025](#)).

For this FYP, the model approach should remain practical. The selected model should:

1. Accept mixed cardiopulmonary audio or its input representation.
2. Estimate separate heart and lung components.
3. Produce audio outputs that can be saved and reviewed.
4. Support evaluation against reference signals when available.
5. Be implementable within the two-trimester FYP timeframe.

6. Be explainable during presentation and viva discussion.

The initial model direction may follow one of three feasible paths:

1. A baseline decomposition method, such as an NMF-style approach, to provide an explainable comparison.
2. A hybrid method that combines signal preprocessing or decomposition with a machine learning model.
3. A U-Net-style or encoder-decoder separation model that maps mixed input representations to separated heart and lung outputs.

More complex methods such as attention networks, phase-aware transformers, or pretrained-model reprogramming are useful as literature references, but they may be too large for a first prototype unless simplified carefully ([Poh et al., 2024](#); [Zhang et al., 2025](#); [Wang et al., 2025](#)). The final method choice will be confirmed after dataset inspection and prototype testing.

3.8 System Architecture Overview

The proposed system architecture is modular. Each module has a clear responsibility so that the prototype can be tested and improved step by step. At a high level, the system will include an input module, preprocessing module, feature/input representation module, separation model module, output management module, evaluation module, and user interaction or interface layer.

The proposed processing flow is:

1. The user provides a mixed cardiopulmonary audio file.
2. The input module validates the file type and stores the input metadata.
3. The preprocessing module converts the audio into a consistent format.
4. The feature extraction or input representation module prepares the data for the model.
5. The separation module estimates heart and lung sound components.
6. The output module reconstructs and stores separated audio files.
7. The evaluation module computes available metrics.
8. The interface or report layer presents the results to the user.

This architecture supports the software-system focus of the project. It avoids treating the model as a standalone script and instead places the model inside a complete audio-processing workflow.

3.9 System Modules

The proposed system modules are:

Module	Purpose	Main input	Main output
Audio input module	Accept and validate mixed cardiopulmonary audio files.	.wav or supported audio file.	Validated input record.
Dataset management module	Organize dataset files, metadata, and train/test splits.	Dataset folders and metadata.	Prepared dataset index.
Preprocessing module	Standardize audio format, duration, amplitude, and segmentation.	Raw audio waveform.	Preprocessed audio segments.
Input representation module	Convert audio into the representation required by the model.	Preprocessed waveform.	Waveform windows, spectrograms, or model tensors.
Separation model module	Estimate separated heart and lung sound components.	Mixed input representation.	Estimated heart and lung representations.
Reconstruction module	Convert model outputs back into playable audio signals.	Estimated source representations.	Heart and lung .wav outputs.
Evaluation module	Compute available separation or signal-quality metrics.	Separated outputs and reference signals where available.	Metric results and evaluation summary.
Output management module	Save separated audio, metadata, and evaluation results.	Audio outputs and metrics.	Organized output files and logs.
User interface or interaction module	Allow user to run separation and review results.	User actions and selected files.	Displayed status, outputs, and metrics.

These modules may be implemented as Python scripts, backend services, or application components depending on the final prototype design.

3.10 Evaluation Methodology

The evaluation methodology will focus on separation quality and system behavior, not disease diagnosis. If the selected dataset provides reference heart and lung source signals, the separated outputs can be compared directly with the references. This is why HLS-CMDS is useful as a dataset candidate: it includes mixed recordings and corresponding source signals for mixed recordings ([Torabi et al., 2025a](#)).

The proposed evaluation will include:

1. Qualitative inspection of separated heart and lung audio outputs.

2. Waveform and spectrogram comparison between mixed input, separated outputs, and reference signals where available.
3. Signal-level error or similarity measures where reference sources are available.
4. Source-separation metrics where supported by the dataset and implementation.
5. Runtime or processing-time observation for the prototype workflow.
6. Documentation of failure cases, noisy samples, and limitations.

The exact metrics will be finalized after confirming the dataset and model. Possible metric categories include signal-to-noise improvement, reconstruction error, correlation or similarity with reference sources, and source-separation quality measures. If reference source signals are not available for some data, the evaluation will be limited to qualitative inspection and indirect signal-quality measures. Classification accuracy will not be used as the main evaluation metric because the project objective is separation rather than disease detection.

3.11 Tools and Technologies

The project will use tools and technologies suitable for Python-based audio processing, machine learning, and software prototyping.

Category	Proposed tools or technologies	Purpose
Programming language	Python	Main implementation language for audio processing and ML modules.
Audio processing	Python audio libraries such as Librosa, SoundFile, SciPy, or similar libraries	Loading audio, resampling, segmentation, waveform processing, and spectrogram generation.
Machine learning	PyTorch, TensorFlow, scikit-learn, or suitable ML libraries	Implementing or adapting the separation model.
Data handling	CSV/JSON metadata files and structured folders	Managing dataset records, input files, outputs, and evaluation summaries.
Visualization	Matplotlib or similar plotting tools	Plotting waveforms, spectrograms, and evaluation outputs.
Software prototype	Python scripts, notebooks, or a lightweight application/backend depending on implementation scope	Running the separation workflow and presenting outputs.
Documentation and project tracking	Markdown notes and project documentation files	Recording methodology, design decisions, and implementation progress.

The final toolset should be confirmed during implementation. Tools should be selected based on project feasibility, reproducibility, and compatibility with the selected dataset and model.

3.12 Chapter Summary

This chapter presented the proposed methodology for the machine learning-based cardiopulmonary sound separation system. The project will follow an iterative software prototyping approach, using public datasets, audio preprocessing, feature or input representation, a feasible machine learning-based separation method, modular system architecture, and separation-focused evaluation.

The methodology supports the proposal by focusing on a software system that separates mixed cardiopulmonary audio into heart and lung sound outputs. It also respects the project boundary by excluding disease detection, clinical diagnosis, and hardware development. The next chapter should develop the system design in more detail, including diagrams, module interactions, data flow, and interface design for the prototype.

4. Chapter 4: Design and Implementation

4.1 Introduction

This chapter presents the design and implementation direction for the machine learning-based cardiopulmonary sound separation system. The chapter translates the methodology from Chapter 3 into a software design that can be implemented as a prototype in the next project phase. The design focuses on audio upload, validation, preprocessing, model-based separation, output generation, result management, and user interaction.

The system is designed as an application-based proof of concept. It is not intended to diagnose heart or lung disease, and it does not make clinical decisions. Its purpose is to demonstrate a structured workflow for separating mixed cardiopulmonary recordings into heart sound and lung sound outputs.

4.2 System Overview

The proposed system accepts a mixed cardiopulmonary audio file and processes it through a sequence of software modules. The user uploads or selects an audio recording, the system validates the file, preprocessing standardizes the audio, and the separation module estimates heart and lung sound components. The separated outputs are saved and made available for review or download. Where reference source signals are available, the system can compute separation-related metrics.

The main user-facing functions are summarized in [Figure 4.1](#). The diagram keeps the scope at system level because the final endpoint names, class names, and implementation details will be confirmed during the implementation phase.

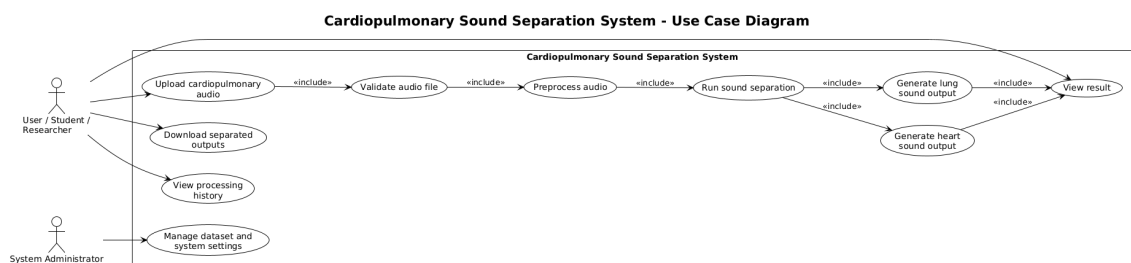


Figure 4.1: Use case diagram for the cardiopulmonary sound separation system.

4.3 System Architecture

The architecture is modular so that each part of the prototype can be implemented, tested, and improved separately. The web interface handles user interaction. The backend API coordinates validation, preprocessing, model inference, result handling, and retrieval of output information. Audio files are

stored as files, while the database stores metadata such as job status, file paths, timestamps, and available metric values. This avoids storing large audio content directly inside the database.

The proposed architecture is shown in [Figure 4.2](#).

Cardiopulmonary Sound Separation System - Component Diagram

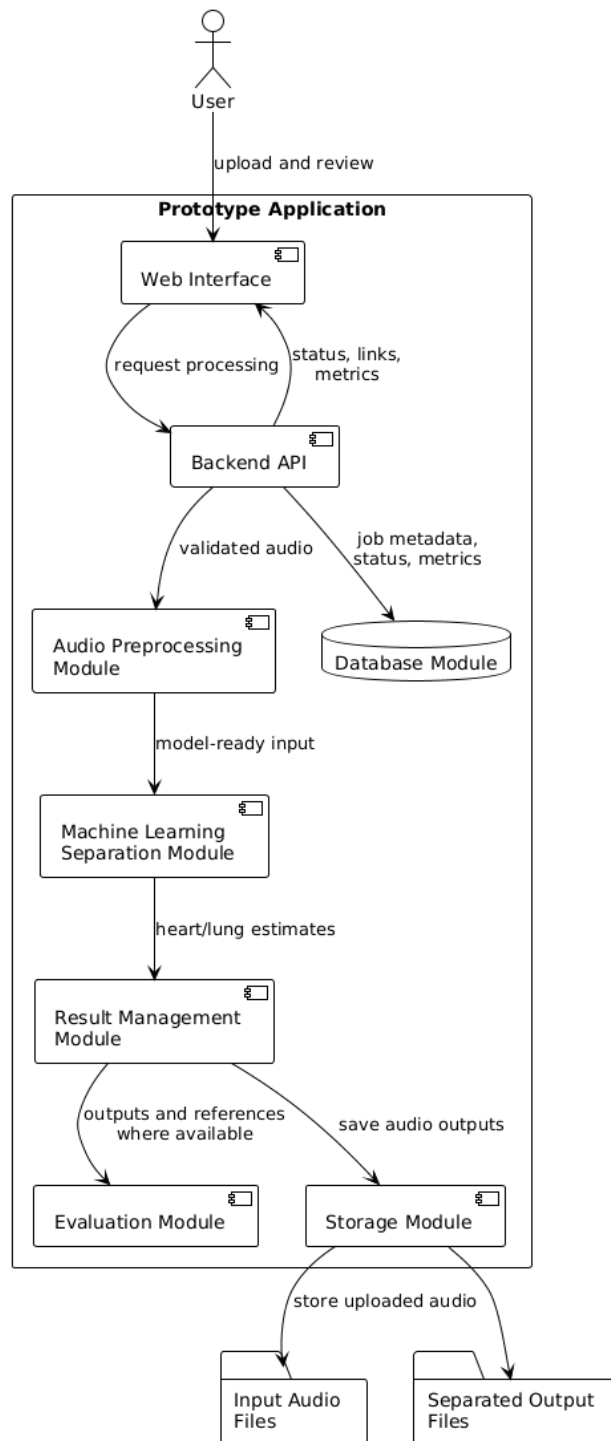


Figure 4.2: Component diagram for the cardiopulmonary sound separation system.

4.4 Main System Modules

The prototype is organized around the modules shown in [Table 4.1](#).

Table 4.1: Main system modules and responsibilities

Module	Main responsibility	Expected output
Web interface	Provides upload, status, result review, and download functions.	User-facing interaction screens.
Backend API	Coordinates validation, preprocessing, model inference, storage, and result retrieval.	Processing responses and result metadata.
Audio preprocessing module	Standardizes sampling rate, channel format, amplitude, and segment length.	Model-ready audio input.
Machine learning separation module	Estimates heart and lung sound components from the mixed recording.	Separated heart and lung estimates.
Result management module	Saves outputs, links metadata, and prepares result summaries.	Organized result records.
Storage module	Stores uploaded and separated audio files.	File paths and stored audio artefacts.
Database module	Stores job metadata, status, output references, and metric values.	Searchable processing history.
Evaluation module	Calculates metrics where reference signals are available.	Metric table or evaluation summary.

4.5 Database and Storage Design

The storage design separates audio files from metadata. Uploaded files and separated heart/lung outputs are stored in structured folders, while the database stores references to these files and information needed to track processing. This approach is more practical for audio processing because .wav files may be large and are easier to manage as files than as database blobs.

The planned database/storage records are summarized in [Table 4.2](#).

Table 4.2: Planned database and storage records

Planned record	Purpose	Example stored information
Audio input record	Tracks each uploaded or selected mixed recording.	Input filename, file path, duration, sampling rate, upload time.
Processing job record	Tracks the status of each separation request.	Job ID, status, preprocessing settings, model used, processing time.
Separation result record	Links the job to generated heart and lung outputs.	Heart output path, lung output path, result timestamp.

Evaluation metric record	Stores metric values when reference signals are available.	SNR, SDR, SI-SDR, MSE, MAE, PESQ, STOI, notes.
System log record	Records important system events and processing errors.	Event type, message, timestamp, related job ID.

At the FYP1 stage, these records describe the intended design. The final database schema may be refined during implementation, but it should preserve the same separation between audio artefacts and metadata.

4.6 Audio Upload and Preprocessing Flow

The audio upload and preprocessing flow begins when the user submits a mixed cardiopulmonary recording. The system checks that the file type, size, and required metadata are valid. A valid file is saved and passed to the preprocessing module. Depending on the selected model, preprocessing may include conversion to mono, resampling, normalization, trimming, padding, segmentation, and time-frequency conversion.

The overall workflow is summarized in [Figure 4.3](#).

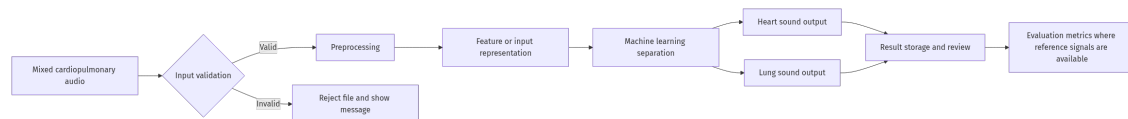


Figure 4.3: Audio processing workflow for the proposed system.

4.7 Machine Learning Model Integration

The machine learning model is integrated as a separate module so that the system can support experimentation with different separation approaches. The model input may be a waveform segment, a spectrogram, or another representation depending on the final implementation. The output should be convertible into two audio signals: one representing the heart sound component and one representing the lung sound component.

The first implementation should prioritize a feasible model that can be trained or adapted using available public data. A baseline or hybrid approach may be used before attempting a more complex model. This design allows the prototype to remain explainable while still supporting improvement in the next phase.

4.8 Heart and Lung Sound Output Generation

After separation, the system reconstructs or saves the estimated heart and lung components as audio outputs. If a time-frequency representation is used, the reconstruction step must convert estimated source representations back into waveform audio. The output generation step should preserve useful metadata

such as sampling rate, duration, model version or method name, and the original input reference.

The expected outputs are:

1. A separated heart sound audio file.
2. A separated lung sound audio file.
3. Processing metadata that links the outputs to the input recording.
4. Evaluation values where reference source signals are available.

4.9 Result Management

Result management handles the storage, retrieval, and presentation of separation outputs. Each processing job should have a clear status such as uploaded, preprocessing, separating, completed, or failed. Completed jobs should provide links to the separated heart and lung sound files and any available evaluation metrics. Failed jobs should provide a clear error message so that the user understands whether the issue came from the file, preprocessing, model inference, or storage.

The expected processing sequence is shown in [Figure 4.4](#).

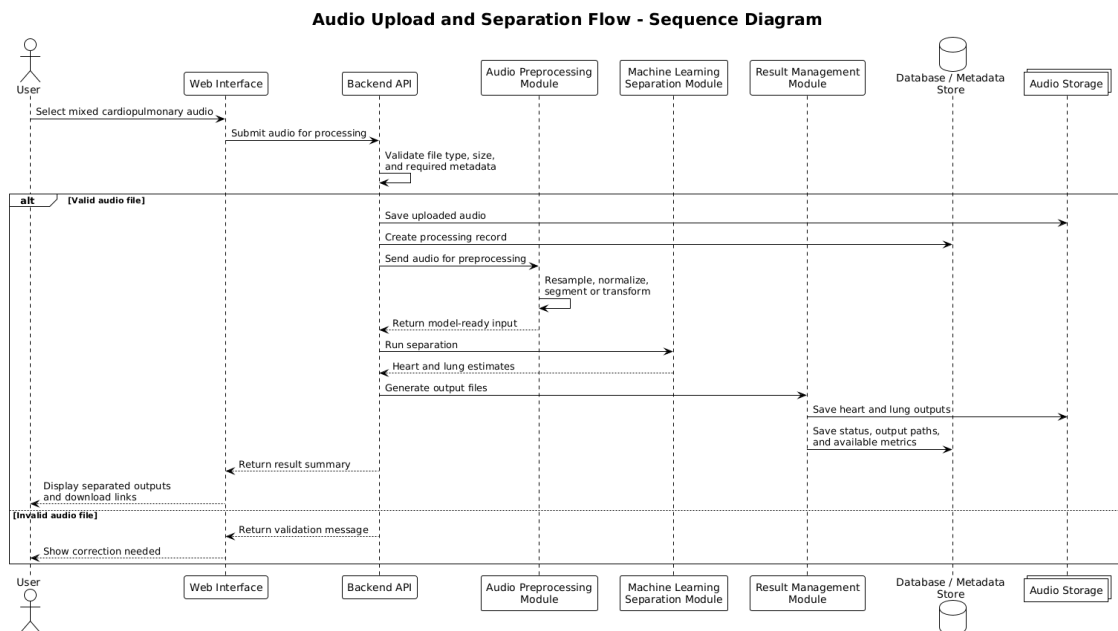


Figure 4.4: Sequence diagram for audio upload and cardiopulmonary sound separation.

4.10 User Interface Design

The user interface should be simple and focused on the main workflow. The first view should allow the user to select or upload a mixed cardiopulmonary audio file. After submission, the interface should show processing status and then display the separated heart and lung outputs when the job is completed. The user should be able to listen to or download the outputs and review any available evaluation metrics.

The planned interface areas are:

1. Audio upload area for selecting a mixed recording.
2. Processing status area for showing validation, preprocessing, and separation progress.
3. Result area for heart sound and lung sound outputs.
4. Metric summary area for available evaluation results.
5. Processing history area for reviewing previous separation jobs if included in the prototype.

The interface should avoid medical wording that implies diagnosis. Labels should describe the outputs as separated heart sound and separated lung sound rather than disease-related results.

4.11 Security, Validation, and Error Handling

The prototype should include basic security and validation controls appropriate for a student software project. File validation should restrict accepted file types and reject corrupted or unsupported audio files. Size limits should prevent very large files from exhausting local storage or memory. Output paths should be generated by the system rather than accepted directly from user input.

Error handling should be clear and traceable. If preprocessing fails, the user should receive a message that the file could not be prepared for processing. If model inference fails, the system should identify the failure as a processing issue rather than producing incomplete outputs. If evaluation cannot be performed because reference signals are unavailable, the system should state that metric calculation is not available for that input instead of reporting unsupported values.

4.12 Implementation Status

This FYP1 report defines the software design, literature foundation, methodology, and planned evaluation approach for the prototype. Full model implementation, training or adaptation, software integration, and dataset-based evaluation will be completed in the next project phase. This status is appropriate

because separation performance should only be reported after the model and evaluation pipeline have been implemented and tested.

The current implementation direction is therefore:

1. Finalize the selected public dataset and prepare the audio files.
2. Implement preprocessing and input representation.
3. Implement or adapt the selected separation method.
4. Integrate the model into the prototype workflow.
5. Generate separated heart and lung sound outputs.
6. Evaluate the outputs using suitable metrics where reference signals are available.
7. Refine the interface and result management based on testing.

4.13 Chapter Summary

This chapter presented the design and implementation direction for the cardiopulmonary sound separation system. The proposed design uses a modular architecture with a web interface, backend API, preprocessing module, machine learning separation module, storage module, database module, result management module, and evaluation module. The chapter also described the planned upload flow, model integration, output generation, user interface, validation, error handling, and implementation status. The next chapter explains how the prototype will be tested and evaluated.

5. Chapter 5: Testing and Evaluation

5.1 Introduction

This chapter presents the testing and evaluation plan for the machine learning-based cardiopulmonary sound separation system. The purpose of testing is to verify that the software prototype can accept mixed cardiopulmonary audio, preprocess the signal, run the selected separation method, generate heart and lung sound outputs, and present the results in a usable form.

At the FYP1 stage, this chapter defines the planned testing strategy and evaluation procedure. Final test results and separation scores should only be reported after the prototype has been fully implemented and evaluated using the selected dataset.

5.2 Testing Strategy

The testing strategy combines functional testing, dataset-based testing, separation metric evaluation, and prototype usability testing. Functional testing checks whether each module behaves correctly. Dataset-based testing checks whether the system can process relevant public cardiopulmonary sound data. Separation evaluation checks the quality of the output signals when reference heart and lung sources are available. Usability testing checks whether the prototype workflow is understandable to a user.

The planned testing will be carried out in stages:

1. Test audio upload and validation with valid and invalid files.
2. Test preprocessing with different file lengths, amplitudes, and sampling rates.
3. Test the separation module using selected mixed recordings.
4. Test output generation for separated heart and lung sound files.
5. Test metric calculation where reference source signals are available.
6. Test result viewing, downloading, and history display if included.
7. Record limitations, failure cases, and improvement actions.

5.3 Functional Testing

Functional testing will verify the main system functions from input to output. The planned functional test cases are shown in [Table 5.1](#).

Table 5.1: Planned functional test cases

Test ID	Function	Test condition	Expected outcome	Status
FT-01	Audio upload	User submits a supported audio file.	File is accepted and a processing record is created.	Planned
FT-02	File validation	User submits an unsupported file type.	System rejects the file and displays a clear validation message.	Planned
FT-03	File validation	User submits a corrupted or unreadable audio file.	System rejects the file without generating output files.	Planned
FT-04	Preprocessing	Valid audio has a different sampling rate or channel format.	Audio is converted to the required processing format.	Planned
FT-05	Separation processing	Valid mixed audio is submitted for separation.	System generates heart and lung sound output files.	Planned
FT-06	Result display	A completed job is opened by the user.	Heart and lung outputs and available metrics are displayed.	Planned
FT-07	Download output	User downloads separated output files.	Downloaded files match the generated heart and lung outputs.	Planned
FT-08	Error handling	Model inference fails or cannot process the input.	System records the failure and shows a meaningful status message.	Planned

5.4 Dataset-Based Testing

Dataset-based testing will use public data that is suitable for cardiopulmonary sound separation. HLS-CMDS is a relevant candidate because it provides heart sounds, lung sounds, mixed recordings, and source signals for mixed recordings ([Torabi et al., 2025a](#)). This makes it suitable for testing both output generation and metric-based evaluation.

The dataset-based testing procedure will include:

1. Select a subset of mixed cardiopulmonary recordings.
2. Confirm that corresponding heart and lung reference signals are available where needed.

3. Run preprocessing on each selected input.
4. Generate separated heart and lung outputs.
5. Compare separated outputs with reference signals using suitable metrics.
6. Record qualitative observations such as audible artefacts, weak separation, or failure cases.

Other public heart sound or lung sound datasets may be used only as supporting data if they fit the test objective. Datasets that do not provide paired mixed and source recordings should not be treated as direct supervised separation evaluation sets.

5.5 Audio Separation Evaluation Metrics

The evaluation metrics should match the separation objective. When reference heart and lung source signals are available, objective signal-level metrics can be computed. When reference signals are not available, evaluation should be limited to qualitative review, waveform or spectrogram inspection, and system behavior observations.

The planned metric categories are summarized in [Table 5.2](#).

Table 5.2: Planned audio separation evaluation metrics

Metric	Purpose	Use condition
SNR	Measures the relationship between useful signal and noise.	Useful when a reference or noise estimate is available.
SDR	Measures distortion in the separated output compared with a reference source.	Suitable for source separation evaluation when references exist.
SI-SDR	Scale-invariant version of SDR that is commonly used in audio separation.	Suitable when output scale differences should not dominate the score.
MSE	Measures mean squared error between separated output and reference.	Useful for reconstruction error analysis.
MAE	Measures average absolute error between separated output and reference.	Useful as a simple reconstruction error measure.
PESQ	Estimates perceptual quality of an audio signal.	May be used cautiously if suitable for the audio type and sampling rate.
STOI	Estimates intelligibility-related quality.	May be supplementary only, because the task is not speech intelligibility.

Processing time	Measures time required for preprocessing and separation.	Useful for prototype performance and usability.
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The final evaluation should not rely on a single metric. A combination of reconstruction error, source-separation metrics, waveform or spectrogram inspection, and processing behavior will provide a more balanced view of prototype performance.

5.6 Evaluation Procedure

The planned evaluation procedure is:

1. Prepare the selected dataset subset and confirm input-reference pairs.
2. Run each mixed recording through the same preprocessing configuration.
3. Apply the selected machine learning-based separation method.
4. Save the separated heart and lung outputs.
5. Align output and reference signals where necessary.
6. Compute supported metrics for each output.
7. Summarize results using tables and visual comparisons.
8. Identify failure cases and explain likely causes.

If the final dataset subset does not provide reference source signals for every input, those samples will be evaluated separately using qualitative inspection and system-level behavior rather than source-separation scores.

5.7 Prototype / Usability Testing

Prototype usability testing will focus on whether the system workflow is clear and usable. This testing is not intended to replace technical evaluation, but it is important because the project is an application-based prototype. The user should be able to upload an audio file, understand whether it was accepted, monitor the processing status, access separated outputs, and identify whether metric values are available.

The planned usability checks are:

1. The upload area is easy to locate and use.
2. Validation messages are clear and specific.
3. Processing status is visible during the workflow.
4. Heart and lung outputs are clearly labelled.
5. Download links or output access options are easy to use.

6. Metric results are presented without implying diagnosis.
7. Error states explain what the user should correct.

5.8 Limitations of Testing

Several limitations are expected. First, the evaluation depends heavily on dataset availability. If a dataset does not provide reference heart and lung source signals, objective separation metrics cannot be computed reliably. Second, manikin-based data may not represent the full variability of real patient recordings, so conclusions should be limited to the data used. Third, the first prototype may prioritize a feasible and explainable model over the most advanced recent architectures. Fourth, usability testing will be limited to prototype workflow review rather than clinical deployment testing.

These limitations do not invalidate the project because the FYP objective is to build and evaluate a software prototype for sound separation. They do, however, define the boundary of what the final results can claim.

5.9 Summary

This chapter presented the planned testing and evaluation approach for the cardiopulmonary sound separation prototype. The testing plan includes functional testing, dataset-based testing, objective audio separation metrics, evaluation procedure, prototype usability checks, and testing limitations. Actual metric scores and final test outcomes will be reported only after implementation and dataset-based evaluation have been completed in the next project phase.

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Appendix A: Full Literature Review Matrix

The complete literature review matrix below summarizes the selected papers used in the Chapter 2 synthesis. It supports the comparison of research focus, methods, datasets, findings, limitations, and relevance to the proposed cardiopulmonary sound separation system.

Table 5.3: Full literature review matrix for selected Chapter 2 papers										
No.	Research Title	Author(s) & Year	Journal / Conference / Database	Research Problem / Objective	Algorithm / Framework Used	Dataset Used	Results / Findings	Research Gap	Limitation	How It Helps This Project
1	An End-to-End Deep Learning Framework for Real-Time Denoising of Heart Sounds for Cardiac Disease Detection in Unseen Noise	Ali et al. (2023) (Ali et al., 2023)	IEEE Access	Improve noisy heart sound recordings before later analysis.	LU-Net encoder-decoder denoiser.	PCG data mixed with breathing and hospital noise.	Reported SNR and real-time denoising improvements.	Denoising is related but does not separate heart and lung outputs.	Focuses on heart sound denoising and diagnosis support.	Supports preprocessing and noise-reduction choices.
2	Noisy Neonatal Chest Sound Separation for High-Quality Heart and Lung Sounds	Grooby et al. (2023) (Grooby et al., 2023)	IEEE Journal of Biomedical and Health Informatics	Separate noisy neonatal chest recordings into heart and lung sounds.	NMF and NMCF.	Neonatal artificial mixtures and noisy chest sounds.	Improved signal quality and vital-sign estimation support.	Robust separation remains difficult under noisy neonatal conditions.	Dataset is neonatal and not clearly public.	Provides a classical separation baseline.
3	NeoSSNet: Real-Time Neonatal Chest Sound Separation Using Deep Learning	Poh et al. (2024) (Poh et al., 2024)	IEEE Open Journal of Engineering in Medicine and Biology	Perform real-time neonatal chest sound separation.	1D encoder-decoder with transformer mask generation.	Neonatal chest recordings with heart and lung references.	Reported SDR and SI-SDR improvement with real-time relevance.	Reproducibility is limited by private clinical data.	Performance depends on noise and recording quality.	Main deep learning reference for the prototype.
4	Recent Advances in PCG Signal Analysis using AI: A Review	Roy et al. (2024) (Roy et al., 2024)	International Journal on Smart Sensing and Intelligent Systems	Review AI methods for PCG preprocessing and analysis.	Review of preprocessing, segmentation, features, and AI models.	PCG datasets from prior studies.	Summarizes common PCG processing stages and AI approaches.	Review is not focused on heart-lung source separation.	Mainly classification and PCG analysis oriented.	Supports background on PCG preprocessing.
5	Research on Heart and Lung Sound Separation Method Based on DAE-NMF-VMD	Sun et al. (2024) (Sun et al., 2024)	EURASIP Journal on Advances in Signal Processing	Improve heart-lung sound separation using a hybrid method.	Deep autoencoder, NMF, and VMD.	Heart-lung sound mixtures.	Reported SDR, PESQ, STOI, and separation metrics.	Reproducible public data access is limited.	Data and code access may require request.	Shows how ML and decomposition can be combined.
6	Unsupervised Framework for Single Channel Heart and Lung Sounds Separation in Data Constrained Environments	Ullah and Zhang (2024) (Ullah and Zhang, 2024)	Applied Acoustics	Separate single-channel heart and lung sounds with limited data.	MCRA, OMLSA, and NMF-HMM.	Heart and lung sound mixtures.	Reported SDR, SAR, and separation performance.	Data-constrained separation needs methods with limited labels.	Dataset availability is unclear.	Supports feasible low-data separation options.
7	A Multi-Channel UNet Framework Based on SNMF-DCNN for Robust Heart-Lung-Sound Separation	Wang et al. (2023) (Wang et al., 2023a)	Computers in Biology and Medicine	Improve robust heart-lung sound separation.	SNMF-DCNN-MCUNet.	Heart-lung mixtures with different lung sound types.	Reported SNR, SDR, SIR, SAR, and correlation improvements.	Constructed mixtures affect generalizability.	Public dataset details are limited.	Guides U-Net-style model design and metrics.
8	Heart-Lung Sound Separation by Nonnegative Matrix Factorization and Deep Learning	Wang et al. (2023) (Wang et al., 2023b)	Biomedical Signal Processing and Control	Combine NMF and deep learning for separation.	MCNMF, clustering, and ECNet.	Heart-lung sound mixtures.	Reported SNR, SDR, SIR, SAR, and correlation.	Separation depends on mixture assumptions.	Public dataset details are limited.	Supports hybrid baseline and evaluation planning.
9	JASSNet: Heart and Lung Sound Separation Network Based on Joint Attention Mechanism	Zhang et al. (2025) (Zhang et al., 2025)	Biomedical Signal Processing and Control	Improve separation using attention and semi-supervision.	Joint attention and semi-supervised network.	Simulated and clinical cardiopulmonary recordings.	Reported SDR, SIR, SAR, and rate-estimation improvement.	Advanced models can be difficult to reproduce.	Clinical data and disease context reduce general reuse.	Defines a recent advanced comparison method.

	and Semi-Supervised Learning									
10	Deep Learning in Heart Sound Analysis: From Techniques to Clinical Applications	Zhao et al. (2024) (Zhao et al., 2024)	Health Data Science	Review deep learning for heart sound analysis.	Review of CNN, RNN, transformer, denoising, and segmentation methods.	Common heart sound datasets.	Summarizes heart sound AI methods and dataset issues.	Not focused on heart-lung source separation.	Clinical analysis focus may not match this project scope.	Supports ML background and dataset discussion.
11	Separation of Cardiopulmonary Sound Signals for Classification of Respiratory Diseases	Zheng and Sanei (2024) (Zheng and Sanei, 2024)	APSIPAASC	Separate cardiopulmonary signals before respiratory analysis.	Singular spectrum analysis with kurtosis optimization.	Auscultated cardiopulmonary recordings.	Separation improved downstream respiratory classification.	Separation quality is secondary to classification.	Diagnostic objective differs from this FYP.	Shows separation value without shifting project scope to diagnosis.
12	Reprogramming Automatic Speech Recognition Models for Neonatal Chest Sound Separation	Poh et al. (2026) (Poh et al., 2026)	IEEE Journal of Biomedical and Health Informatics	Adapt speech models for neonatal chest sound separation.	Whisper model reprogramming.	Neonatal chest mixtures and supporting PCG data.	Reported separation, segmentation, and rate-estimation results.	Foundation-model adaptation is complex.	Small neonatal source set and artificial mixtures limit generalization.	Identifies an advanced ML direction beyond the first prototype.
13	A Phase-Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation	Wang et al. (2025) (Wang et al., 2025)	IEEE Journal of Biomedical and Health Informatics	Improve single-channel chest sound reconstruction.	Phase-enhanced dual-path transformer.	Constructed heart and lung sound mixtures.	Reported SI-SDR-based and phase-aware improvements.	Phase reconstruction is often under-addressed.	Relies on constructed mixtures.	Supports time-frequency and reconstruction design choices.
14	Large Language Model-based Nonnegative Matrix Factorization For Cardiorespiratory Sound Separation	Torabi et al. (2025) (Torabi et al., 2025b)	arXiv	Use language-model feedback to tune NMF separation.	LingoNMF with LLaMA-based feedback.	Synthetic and manikin heart-lung recordings.	Reported SNR, SDR, SIR, and SAR improvements.	LLM-assisted separation adds system complexity.	Preprint and synthetic/manikin data limit claims.	Helps compare explainable NMF extensions.
15	A New Non-Negative Matrix Factorization Approach for Blind Source Separation of Cardiovascular and Respiratory Sound Based on the Periodicity of Heart and Lung Function	Torabi et al. (2023) (Torabi et al., 2023)	arXiv	Use periodicity to separate cardiovascular and respiratory sounds.	Periodicity-guided NMF.	Synthetic heart and lung sound mixtures.	Reported SDR, SIR, and SAR.	Real-world validation remains limited.	Mainly synthetic evaluation.	Supports physiological periodicity as a separation cue.
16	Separation of Heart and Lung Sounds by A Deep Network-Based Model	Isai et al. (2023) (Isai et al., 2023)	International Conference on Knowledge Innovation and Invention	Separate generated heart-lung mixtures using a deep model.	Deep network-based model.	Public heart and lung sounds used to generate mixtures.	Reported correlation-based separation results.	Evaluation needs stronger metrics and real recordings.	Synthetic mixtures and limited method detail.	Provides a feasible student-level ML reference.
17	An efficient lung sound separation algorithm base on GIHO-VMD	Zhang et al. (2026) (Zhang et al., 2026)	Computer Methods and Programs in Biomedicine	Separate lung sounds from mixed heart-lung signals.	GIHO-VMD.	Pre-simulated mixed heart and lung sound signals.	Reported SNR, NCC, spectrogram, and runtime comparisons.	Decomposition methods remain sensitive to simulation assumptions.	Real-world application pipeline is not fully shown.	Provides a recent decomposition comparator.
18	Identifying the Respiratory Sound Based on Single-Channel Separation and Hyperdimensional Computing	Zheng et al. (2025) (Zheng et al., 2025)	IEEE Sensors Journal	Use separation before respiratory sound identification.	Encoder-decoder separation, hyperdimensional computing, and LightGBM.	ICU respiratory recordings and benchmark data.	Reported respiratory identification improvements.	Separation is used mainly for classification.	Standalone separation quality is not the main objective.	Shows how separation can support later audio analysis.
19	Descriptor: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope (HLS-CMDS)	Torabi et al. (2025) (Torabi et al., 2025a)	IEEE Data Descriptions	Provide paired recordings for separation research.	Dataset descriptor.	HLS-CMDS manikin heart, lung, mixed, and source recordings.	Provides mixed recordings with corresponding source signals.	Public paired source datasets remain limited.	Manikin data may not represent full patient variability.	Main public dataset candidate for training and evaluation.
20	Automatic Wheeze Segmentation Using Harmonic-Percussive Source Separation and Empirical Mode Decomposition	Rocha et al. (2023) (Rocha et al., 2023)	IEEE Journal of Biomedical and Health Informatics	Segment wheezes from respiratory recordings.	HPSS and EMD.	Respiratory sound datasets.	Reported wheeze segmentation performance.	Respiratory preprocessing is not heart-lung separation.	Full real-world wheeze segmentation remains difficult.	Supports preprocessing and respiratory signal handling.

21	A Blind Filtering Framework for Noisy Neonatal Chest Sounds	Fattahi et al. (2022) (Fattahi et al., 2022)	IEEE Access	Filter noisy neonatal chest sounds.	Wavelet transform, PCA, periodic component analysis, and SOBI.	Neonatal chest sound recordings.	Improved quality assessment and rate estimation.	Traditional filtering needs careful component selection.	Severe artefacts may still reduce reliability.	Supports preprocessing and baseline separation design.
22	A Dual Classifier-Regressor Architecture for Heart Sound Onset/Offset Detection	Somarathne et al. (2026) (Somarathne et al., 2026)	IEEE Transactions on Biomedical Engineering	Detect heart sound onset and offset.	Classifier-regressor PCG architecture.	PhysioNet/CinC heart sound data.	Reported timing and segmentation metrics.	Segmentation is not source separation.	Requires PCG/ECG-style context.	Supports timing and evaluation considerations.
23	BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems	Ali et al. (2026) (Ali et al., 2026)	Computer Methods and Programs in Biomedicine Update	Provide a public heart sound dataset.	Dataset descriptor.	BUET multi-disease heart sound recordings.	Provides labelled heart sound data for AI research.	Dataset is diagnosis-oriented, not paired separation data.	No lung source or mixed-source separation target.	Helps distinguish background datasets from separation datasets.
24	Cardiorespiratory Sound Separation Using Singular Spectrum Analysis	Han and Quan (2025) (Han and Quan, 2025)	International Conference on Signal Processing Systems Sensors	Separate cardiac and respiratory components.	Multi-stage singular spectrum analysis.	Digital stethoscope cardiorespiratory recordings.	Presents a recent direct separation approach.	Detailed evaluation evidence remains limited at this reporting stage.	Full implementation details need further confirmation.	Adds a recent non-deep-learning comparator.
25	StethAid: A Digital Auscultation Platform for Pediatrics	Arjouni et al. (2023) (Arjouni et al., 2023)	Sensors	Support pediatric digital auscultation workflow.	Digital auscultation platform.	Pediatric auscultation use context.	Demonstrates platform-oriented auscultation support.	Platform design is not separation-specific.	Does not provide heart-lung separation output.	Supports software workflow and interface thinking.
26	Sound-Dr: Reliable Sound Dataset and Baseline Artificial Intelligence System for Respiratory Illnesses	Hoang et al. (2023) (Hoang et al., 2023)	PHM Society Asia-Pacific Conference	Provide respiratory audio dataset and baseline AI.	Dataset and baseline AI system.	Sound-Dr respiratory recordings.	Supports respiratory sound dataset context.	Respiratory illness focus differs from source separation.	No paired heart-lung source targets.	Provides supplementary dataset background.
27	Compressed Sensing of Acoustic Cardiopulmonary Signals Using a CNN-based Reconstruction Method	Baeyens et al. (2025) (Baeyens et al., 2025)	IEEE EMBC	Reconstruct acoustic cardiopulmonary signals.	Compressed sensing with CNN reconstruction.	Acoustic cardiopulmonary signals.	Shows reconstruction quality can affect later processing.	Reconstruction is not heart-lung source separation.	Dataset and deployment details need confirmation.	Supports acquisition and reconstruction context.
28	Real-Time Multi-Level Neonatal Heart and Lung Sound Quality Assessment for Telehealth Applications	Grooby et al. (2022) (Grooby et al., 2022)	IEEE Access	Assess neonatal heart and lung sound quality.	Real-time multi-level quality assessment.	Neonatal heart and lung sound recordings.	Supports input quality assessment before analysis.	Quality assessment is separate from source separation.	Does not generate separated outputs.	Supports input-quality checks for the prototype.
29	Bridging Auscultation and Tiny Machine Learning: A Digital Stethoscope Leveraging Convolutional Neural Networks on an Embedded Device for Organ Sound Analysis	Mutlu et al. (2024) (Mutlu et al., 2024)	Traitement du Signal	Process organ sounds on constrained devices.	CNN-based TinyML digital stethoscope system.	Organ sound recordings.	Demonstrates embedded auscultation processing.	Organ classification is not source separation.	Hardware direction is outside this software prototype.	Provides implementation context for practical systems.
30	Heart murmur detection from phonocardiogram recordings: The George B. Moody PhysioNet Challenge 2022	Reyna et al. (2023) (Reyna et al., 2023)	PLOS Digital Health	Benchmark heart murmur detection from PCG.	Challenge dataset and scoring framework.	CinC/DigiScope pediatric PCG data.	Provides benchmark context for heart sound analysis.	Detection task is not source separation.	Does not include lung source separation targets.	Supports dataset and evaluation background.
31	A robust deep learning based model for denoising phonocardiogram signals in clinical environments	Jakubec et al. (2025) (Jakubec et al., 2025)	Engineering Applications of Artificial Intelligence	Denoise PCG signals in noisy environments.	Deep learning denoising model.	PCG datasets.	Supports robust PCG noise reduction.	Denoising does not produce both heart and lung outputs.	Dataset and metric details require later confirmation.	Supports preprocessing design.

32	A self-attention-driven deep denoiser model for real time lung sound denoising in noisy environments	Shuvo et al. (2026) (Shuvo et al., 2026)	Biomedical Signal Processing and Control	Denoise lung sounds in real time.	Self-attention deep denoiser.	Noisy lung sound recordings.	Shows attention-based denoising potential.	Lung denoising is not heart-lung separation.	Does not output separated cardiac sound.	Supports respiratory preprocessing choices.
33	Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds	Behera et al. (2026) (Behera et al., 2026)	Biomedical Signal Processing and Control	Denoise real-life respiratory recordings.	DWT coefficient thresholding with UNet.	Real-life respiratory sounds.	Supports multiscale denoising for noisy respiratory data.	Focuses on respiratory denoising rather than source separation.	No paired heart-lung outputs.	Supports wavelet and UNet preprocessing ideas.
34	Lung sound signal denoising using discrete wavelet transform and artificial neural network	Pouyani et al. (2022) (Pouyani et al., 2022)	Biomedical Signal Processing and Control	Improve noisy lung sound quality.	DWT and artificial neural network.	Lung sound signals.	Supports classical and ML-based lung denoising.	Does not address cardiopulmonary source separation.	Dataset details need confirmation.	Supports preprocessing and denoising background.
35	Automatic breathing phase identification based on the second derivative of the recorded lung sounds	Pal et al. (2024) (Pal and Barney, 2024)	Biomedical Signal Processing and Control	Identify breathing phases from lung sounds.	Second-derivative based phase identification.	Heterogeneous lung sound recordings.	Reported breathing phase identification metrics.	Phase identification is not source separation.	Focuses only on respiratory signal structure.	Supports lung sound segmentation and evaluation context.

Appendix B: PRISMA Screening Summary

The PRISMA screening process used in Chapter 2 is summarized below. The summary separates eligible papers used in the literature synthesis from eligible backup records that were not cited in the current report.

Table 5.4: PRISMA screening summary for the literature review

Screening item	Count	Report interpretation
Records considered	96	Records reviewed for relevance to cardiopulmonary sound separation and supporting topics.
Records excluded	47	Records removed because they were outside scope, inaccessible for supporting use, duplicated, or weakly related to the project.
Eligible records after screening	49	Records remaining after screening and access checking.
Studies selected for Chapter 2 synthesis	35	Papers used in the Chapter 2 narrative, summary matrix, or full matrix.
Core separation studies	18	Studies directly related to cardiopulmonary, heart-lung, or chest sound separation.
Background/supplementary studies	17	Studies related to datasets, preprocessing, machine learning methods, evaluation metrics, or supporting context.
Eligible backup records not cited	14	Eligible records retained for future reference but not used as evidence in this report.

Appendix C: System Design Diagrams

The main system design diagrams are presented in Chapter 4. They summarize the prototype use cases, component structure, audio processing sequence, and workflow. The diagrams are included to support the design explanation without adding unsupported implementation details.

Table 5.5: System design diagram summary

Diagram	Purpose
Use case diagram	Shows the user-facing functions of the cardiopulmonary sound separation system.
Component diagram	Shows the main software modules and how they interact.
Sequence diagram	Shows the expected processing flow when a user uploads an audio file.
Audio processing workflow	Summarizes validation, preprocessing, separation, output generation, and evaluation.

Appendix D: Planned Test Cases and Evaluation Metrics

The following planned tests will guide the FYP2 implementation and evaluation phase. Actual results should be recorded only after the prototype has been implemented and tested.

Table 5.6: Appendix planned test cases and evaluation checks

Test area	Planned check	Expected evidence
Audio upload	Accept a valid cardiopulmonary audio file.	Successful upload status and saved input metadata.
File validation	Reject unsupported file type, missing file, or oversized file.	Clear validation message without processing failure.
Preprocessing	Convert audio to the required sampling rate, channel format, and segment length.	Stored preprocessing metadata and consistent model input.
Model inference	Run the selected separation method on a valid mixed input.	Generated heart and lung output files.
Output review	Allow separated outputs to be played, inspected, or downloaded.	Accessible result view and downloadable audio files.
Metric calculation	Compute relevant metrics when reference source signals are available.	Evaluation table containing valid metric values.
Error handling	Handle corrupted or invalid audio input safely.	User-facing error message and no corrupted output record.